GenAI Unit 2 – Assignment 2 (MoE)

Name: K L Sonika **SRN:** PES2UG23CS247

Section: D **Date:** 24-02-2026

2. Environment Variable Management

Load `.env`, ensure `GROQ_API_KEY` exists, echo a masked state, and instantiate a reusable Groq client.

```python
import os
from pathlib import Path
from getpass import getpass
from dotenv import load_dotenv
from groq import Groq

ENV_PATH = Path(".env")
load_dotenv(ENV_PATH, override=False)

GROQ_API_KEY = os.getenv("GROQ_API_KEY")
if not GROQ_API_KEY:
    GROQ_API_KEY = getpass("Enter your GROQ API key: ").strip()
    if not GROQ_API_KEY:
        raise EnvironmentError(
            "Missing GROQ_API_KEY. Set it in .env or provide it interactively when prompted."
        )

masked_key = f"{'*' * 6}{GROQ_API_KEY[-4:]}"
print(f".env located at: {ENV_PATH.resolve()}")
print(f"GROQ_API_KEY loaded: {masked_key}")

groq_client = Groq(api_key=GROQ_API_KEY)
```

[3] ✓ 0.3s

```
.env located at: C:\Users\klson\OneDrive\Desktop\GenAI Assignment\.env
GROQ_API_KEY loaded: ******Lxic
```

```python
    },
    "general": {
        "model": BASE_MODEL,
        "temperature": 0.7,
        "top_p": 0.95,
        "max_tokens": 380,
        "system_prompt": (
            "You are General Expert MoE-03, a friendly concierge for lightweight or sales-style chats. "
            "Keep answers concise, upbeat, and redirect to technical or billing when appropriate."
        ),
    },
    "tool_use": {
        "model": BASE_MODEL,
        "temperature": 0.2,
        "top_p": 0.5,
        "max_tokens": 60,
        "system_prompt": (
            "You are Tool Expert MoE-04 who prefers calling deterministic tools for data lookups. "
            "Respond with 'tool_use' when you need an external capability."
        ),
    },
}

print(f"Configured experts: {list(MODEL_CONFIG.keys())}")
```

[4] ✓ 0.0s

```
Configured experts: ['technical', 'billing', 'general', 'tool_use']
```

4. Router Prompt Design and Implementation

Set up logging plus a deterministic router that calls Groq's `llama-3.3-70b-versatile` model at temperature 0 to emit only the category name.

```python
import logging

logger = logging.getLogger("moe_router")
if not logger.handlers:
    handler = logging.StreamHandler()
    formatter = logging.Formatter("%(asctime)s - %(name)s - %(levelname)s - %(message)s")
    handler.setFormatter(formatter)
    logger.addHandler(handler)

logger.setLevel(logging.INFO)
logger.propagate = False

print("Logger ready:", logger)
```

[5] ✓ 0.0s

```
Logger ready: <Logger moe_router (INFO)>
```

```python
        verdict = "PASS" if predicted == expected else "WARN"
        results.append((text, expected, predicted, verdict))
        print("-" * 60)
        print(f"Input: {text}")
        print(f"Expected: {expected} | Predicted: {predicted} | Verdict: {verdict}")

    if any(item[3] != "PASS" for item in results):
        print("\nOne or more cases need attention. Review WARN/ERROR rows above.")
    else:
        print("\nAll routing checks passed.")
```

[9] ✓ 0.3s Python

```
2026-02-24 23:01:01,145 - moe_router - INFO - Router selected category='technical' for text='My Python script is throwing an IndexError on line 5.'
2026-02-24 23:01:01,250 - moe_router - INFO - Router selected category='billing' for text='I was charged twice for my subscription this month.'
2026-02-24 23:01:01,341 - moe_router - INFO - Router selected category='general' for text='Can you tell me about upcoming product webinars?'
2026-02-24 23:01:01,342 - moe_router - INFO - Keyword match triggered direct 'tool_use' routing.
------------------------------------------------------------
Input: My Python script is throwing an IndexError on line 5.
Expected: technical | Predicted: technical | Verdict: PASS
------------------------------------------------------------
Input: I was charged twice for my subscription this month.
Expected: billing | Predicted: billing | Verdict: PASS
------------------------------------------------------------
Input: Can you tell me about upcoming product webinars?
Expected: general | Predicted: general | Verdict: PASS
------------------------------------------------------------
Input: What is the current price of Bitcoin right now?
Expected: tool_use | Predicted: tool_use | Verdict: PASS

All routing checks passed.
```

[10] ✓ 2.4s Python

```
2026-02-24 23:01:01,444 - moe_router - INFO - Router selected category='technical' for text='My python script is throwing an IndexError on line 5.'
============================================================
User: My python script is throwing an IndexError on line 5.
2026-02-24 23:01:02,609 - moe_router - INFO - Expert 'technical' responded (usage=CompletionUsage(completion_tokens=512, prompt_tokens=80, total_tokens=592,
2026-02-24 23:01:02,716 - moe_router - INFO - Router selected category='billing' for text='I was charged twice for my subscription this month.'
Response: To provide a more accurate diagnosis, I'll need to see the code snippet that's causing the issue. However, I can guide you through a general apprc

An `IndexError` typically occurs when you're trying to access an element in a sequence (like a list, tuple, or string) using an index that's out of range. F

Here's

============================================================
User: I was charged twice for my subscription this month.
2026-02-24 23:01:03,380 - moe_router - INFO - Expert 'billing' responded (usage=CompletionUsage(completion_tokens=199, prompt_tokens=78, total_tokens=277, c
2026-02-24 23:01:03,466 - moe_router - INFO - Router selected category='general' for text='Can you tell me about upcoming product webinars?'
Response: I'm so sorry to hear that you were charged twice for your subscription this month. I can imagine how frustrating that must be for you. Please know

Our standard operating procedure for handling duplicate charges is to process a refund for the incorrect charge within 3-5 business days of receiving your r

============================================================
User: Can you tell me about upcoming product webinars?
2026-02-24 23:01:03,801 - moe_router - INFO - Expert 'general' responded (usage=CompletionUsage(completion_tokens=67, prompt_tokens=80, total_tokens=147, cc
2026-02-24 23:01:03,802 - moe_router - INFO - Keyword match triggered direct 'tool_use' routing.
C:\Users\klson\AppData\Local\Temp\ipykernel_21464\952318334.py:6: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in
  timestamp = datetime.utcnow().isoformat(timespec="seconds")
2026-02-24 23:01:03,803 - moe_router - INFO - Tool-use expert responded via deterministic helper.
Response: We have some exciting webinars coming up. For the most up-to-date schedule and to register, I recommend checking our website or reaching out to ou

============================================================
User: Give me the current price of Bitcoin.
Response: Tool-Use Expert Report
Mock BTC price: $63250.42 (UTC 2026-02-24T17:31:03)
Reminder: Replace this mock with a live market data integration when ready.
```